Mobile Application Development

Practical No. 12

Develop a simple calculator which uses grid layout and GUI concepts.

Que. 1 - Design a calculator UI using GridLayout with buttons for digits and basic operators.

Que. 2 - Implement logic to perform addition, subtraction, multiplication, and division.

Que. 3 - Add a clear button to reset the input and result.

Que. 4 - Display the result dynamically in a TextView after pressing "=".

Que. 5 - Modify the calculator to support decimal inputs and operations.

Que. 6 - Add error handling for invalid inputs or divide-by-zero cases.

ANS:-

1 - activity_main.xml

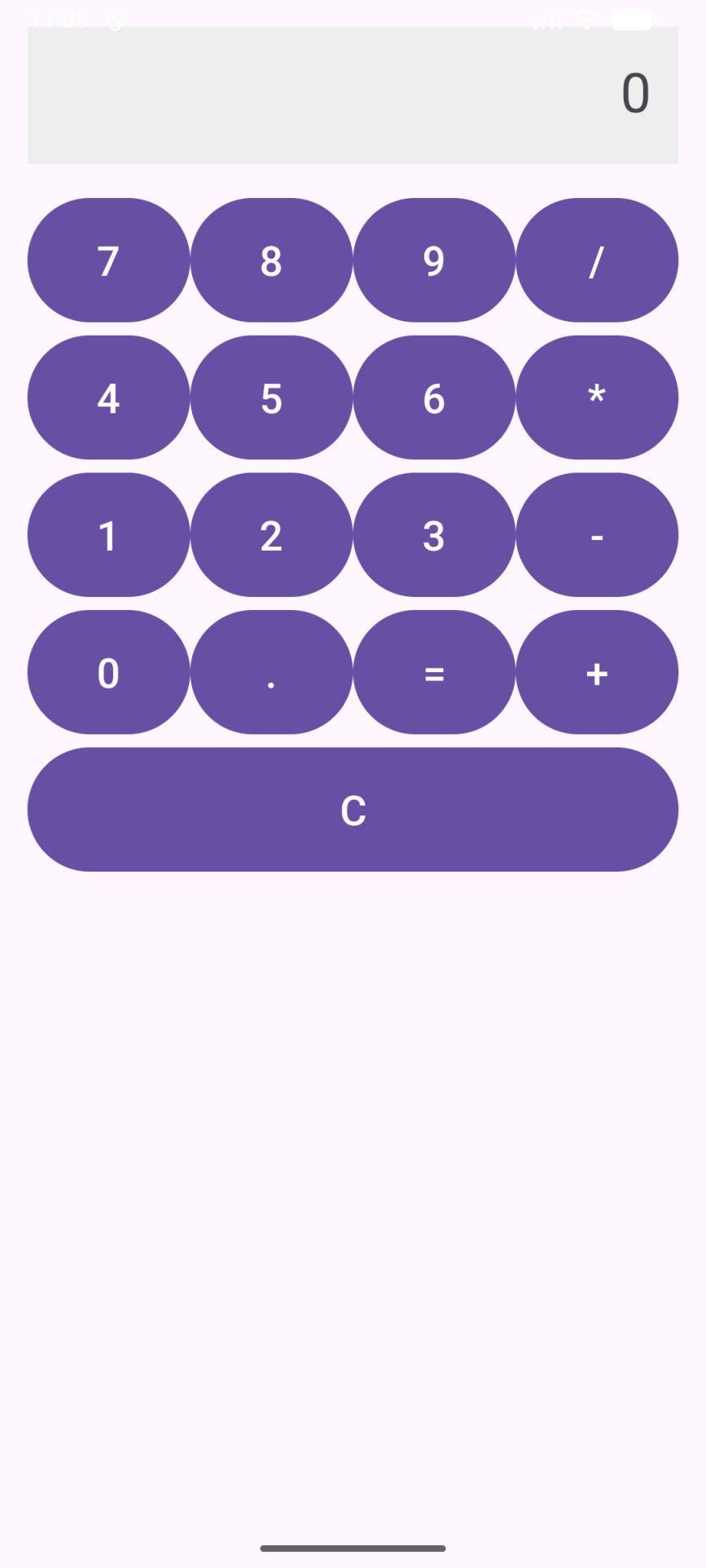
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Display -->
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="80dp"
        android:gravity="end"
        android:textSize="32sp"
        android:background="#EEEEEE"
        android:padding="16dp"
        android:text="0"/>

    <!-- GridLayout for Buttons -->
    <GridLayout
        android:id="@+id/gridLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:columnCount="4"
        android:rowCount="5"
        android:layout_marginTop="16dp">

        <!-- Row 1 -->
        <Button android:text="7" style="@style/CalcButton"/>
        <Button android:text="8" style="@style/CalcButton"/>
        <Button android:text="9" style="@style/CalcButton"/>
        <Button android:text="/" style="@style/CalcButton"/>

        <!-- Row 2 -->
```

```
        <Button android:text="4" style="@style/CalcButton"/>
        <Button android:text="5" style="@style/CalcButton"/>
        <Button android:text="6" style="@style/CalcButton"/>
        <Button android:text="*" style="@style/CalcButton"/>

        <!-- Row 3 -->
        <Button android:text="1" style="@style/CalcButton"/>
        <Button android:text="2" style="@style/CalcButton"/>
        <Button android:text="3" style="@style/CalcButton"/>
        <Button android:text="-" style="@style/CalcButton"/>

        <!-- Row 4 -->
        <Button android:text="0" style="@style/CalcButton"/>
        <Button android:text="." style="@style/CalcButton"/>
        <Button android:text="=" style="@style/CalcButton"/>
        <Button android:text="+" style="@style/CalcButton"/>

        <!-- Row 5 -->
        <Button android:text="C"
            android:layout_columnSpan="4"
            style="@style/CalcButton"/>
    </GridLayout>

 </LinearLayout>
```

2 – mainActivity.java

```
package com.example.cal;

import androidx.appcompat.app.AppCompatActivity;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.GridLayout;

public class MainActivity extends AppCompatActivity {

    TextView tvResult;
    String currentInput = "";
    double firstNumber = 0;
    String operator = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvResult = findViewById(R.id.tvResult);
        @SuppressLint({"MissingInflatedId", "LocalSuppress"}) GridLayout
gridLayout = findViewById(R.id.gridLayout);

        for(int i = 0; i < gridLayout.getChildCount(); i++) {
            View view = gridLayout.getChildAt(i);
            if(view instanceof Button) {
                ((Button) view).setOnClickListener(this::onButtonClick);
            }
        }
    }

    private void onButtonClick(View view) {
        Button button = (Button) view;
        String value = button.getText().toString();
```

```
        switch (value) {

            case "C":
                currentInput = "";
                firstNumber = 0;
                operator = "";
                tvResult.setText("0");
                break;

            case "+":
            case "-":
            case "*":
            case "/":
                if (!currentInput.isEmpty()) {
                    firstNumber = Double.parseDouble(currentInput);
                    operator = value;
                    currentInput = "";
                }
                break;

            case "=":
                calculateResult();
                break;

            default:
                currentInput += value;
                tvResult.setText(currentInput);
        }
    }

    private void calculateResult() {
        if (currentInput.isEmpty() || operator.isEmpty()) {
            Toast.makeText(this, "Invalid Input", Toast.LENGTH_SHORT).show();
            return;
        }

        double secondNumber = Double.parseDouble(currentInput);
        double result = 0;

        try {
            switch (operator) {
                case "+":
                    result = firstNumber + secondNumber;
                    break;
                case "-":
                    result = firstNumber - secondNumber;
                    break;
                case "*":
                    result = firstNumber * secondNumber;
                    break;
                case "/":
                    if (secondNumber == 0) {
                        tvResult.setText("Error");
                        return;
                    }
                    result = firstNumber / secondNumber;
                    break;
            }

            tvResult.setText(String.valueOf(result));
            currentInput = String.valueOf(result);
            operator = "";

        } catch (Exception e) {
            tvResult.setText("Error");
        }
```

```
        }
}
```

Output:-